

Swarmfront Analytics Roadmap

Core Philosophy

Analytics = structured observations of SimState over time. Not UI events. Not guesses. Truth only.

Layer Architecture

Layer A — Raw Signals (engine)

Layer B — Metrics (math)

Layer C — Insights (logic / AI)

Tier 1 — Feel & Mechanics

- Reaction time
- Open budget time (idle %)
- Actions per minute (meaningful only)
- Early game activity score

Tier 2 — Efficiency Engine

- Units produced
- Units sent
- Units landed
- Waste ratio
- Lane utilization %

Tier 3 — Decision DNA

- Aggression index
- Focus vs spread
- Overextension rate
- Conversion efficiency

- Timing discipline

Tier 4 — AI Coaching (Boss Mode)

- Identify turning point
- Highlight inefficiencies
- Provide 1–3 actionable improvements

Core Signals (Engine-Level)

- Action timestamps
- Unit counts
- Resource levels
- Ownership changes
- Lane actions

Implementation Plan

1. Define event schema
2. Add hooks in SimState
3. Log JSON per match
4. Review manually and iterate